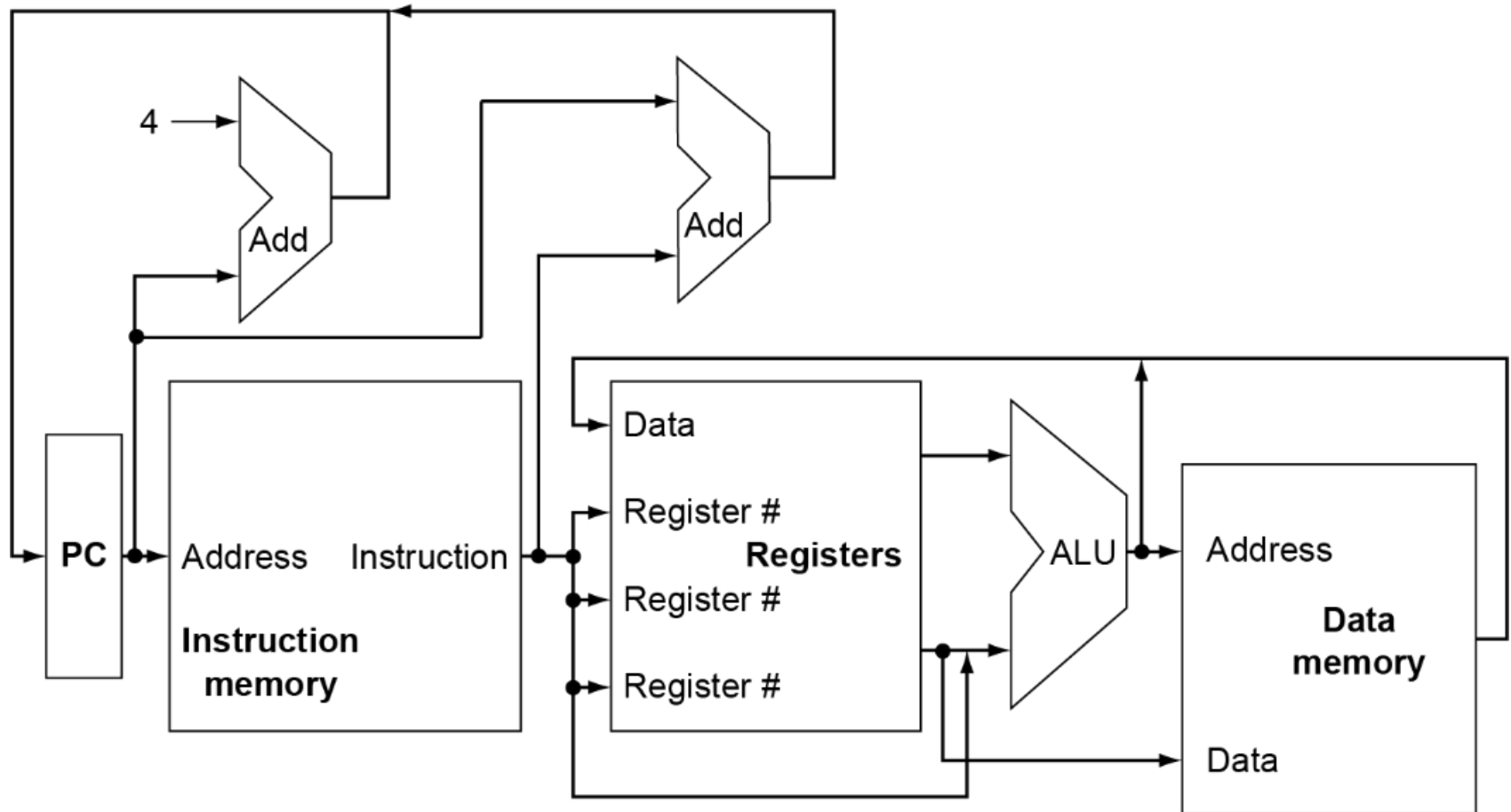


Chapter 4

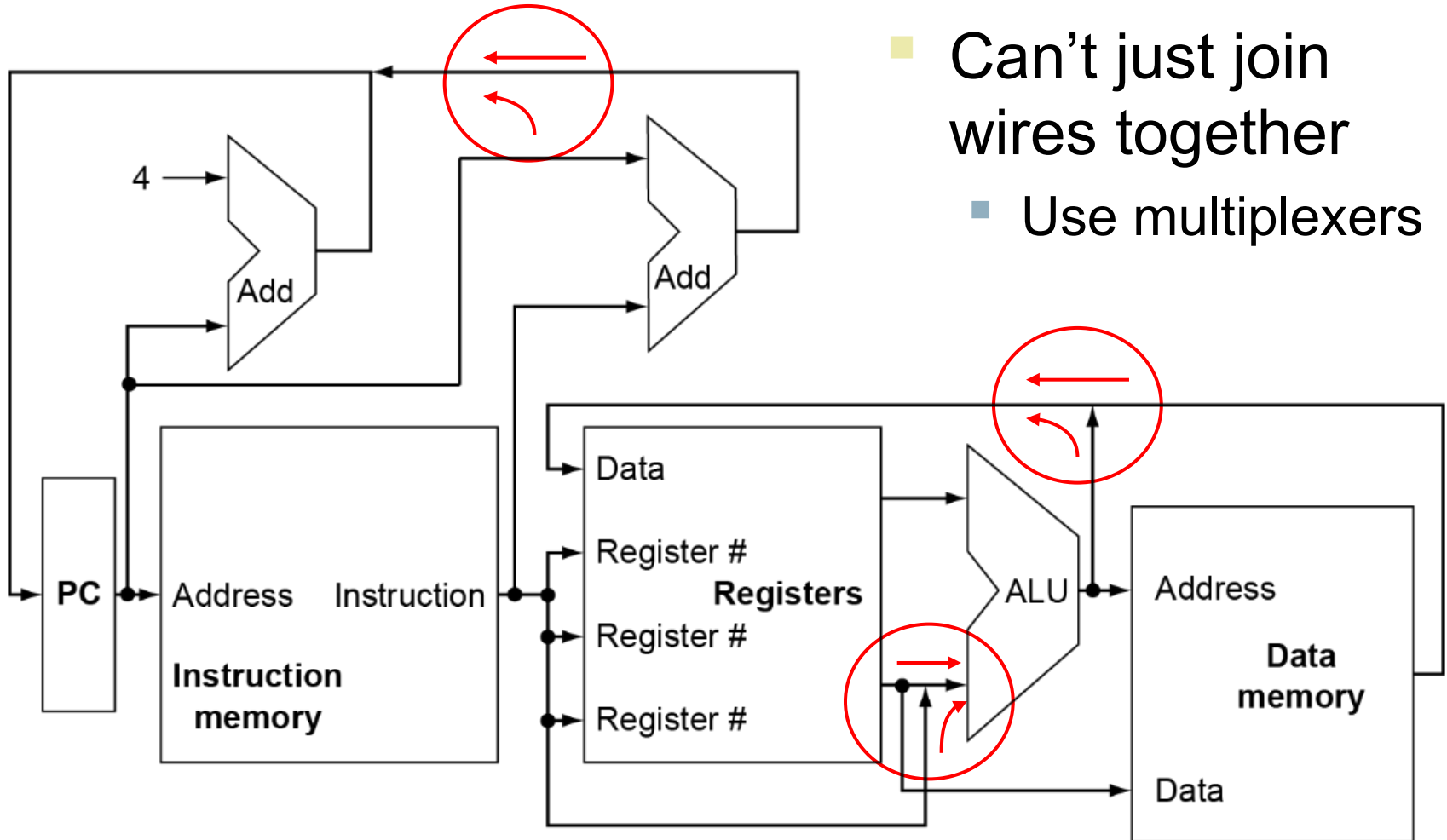
The Processor

CPU Overview

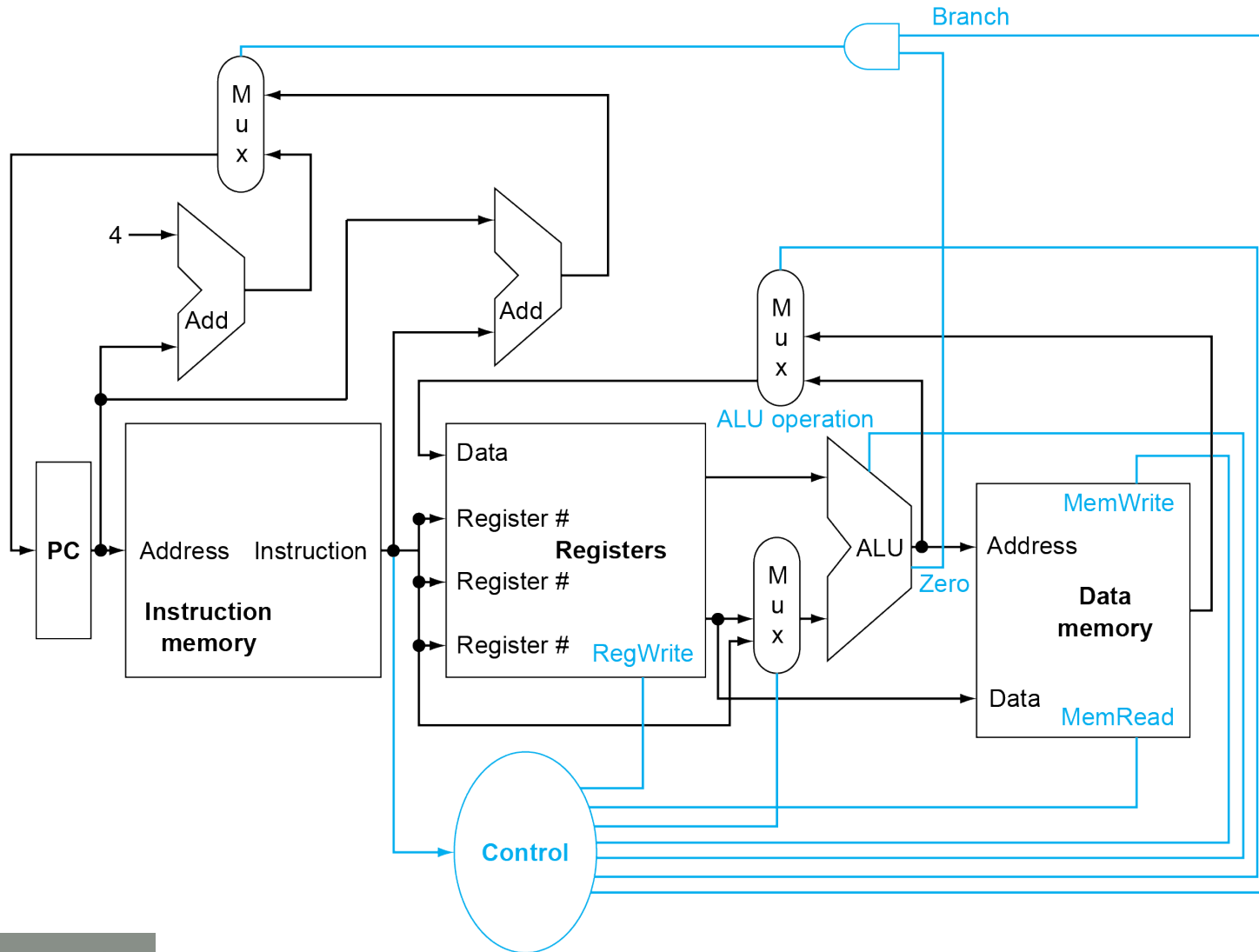


Multiplexers

- Can't just join wires together
 - Use multiplexers



Control



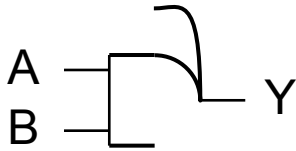
Logic Design Basics

- Information encoded in binary
 - Low voltage = 0, High voltage = 1
 - One wire per bit
 - Multi-bit data encoded on multi-wire buses
- Combinational element
 - Operate on data
 - Output is a function of input
- State (sequential) elements
 - Store information

Combinational Elements

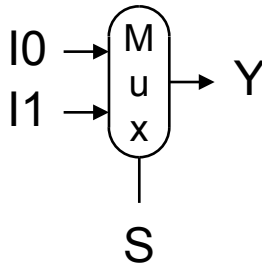
- AND-gate

- $Y = A \& B$



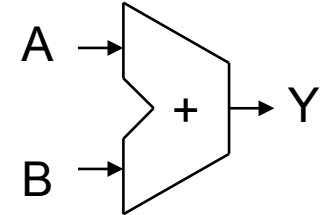
- Multiplexer

- $Y = S ? I1 : I0$



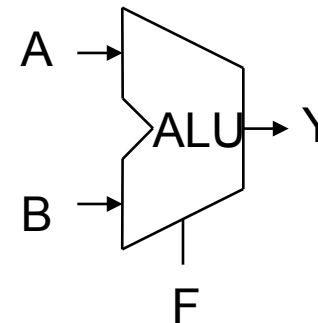
- Adder

- $Y = A + B$



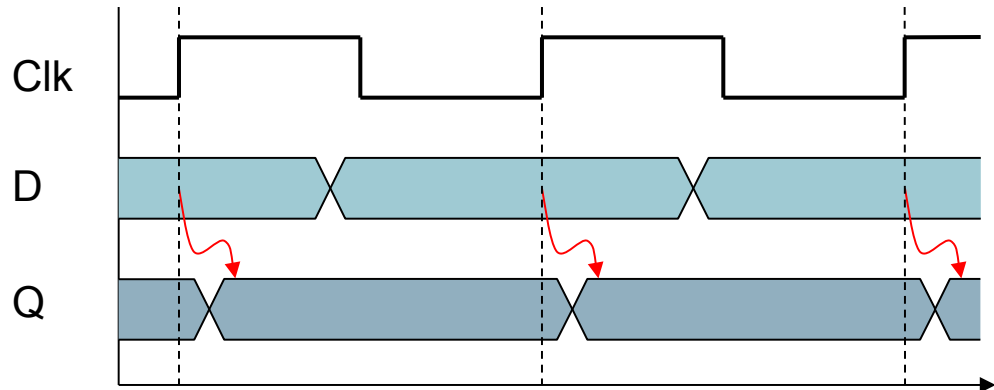
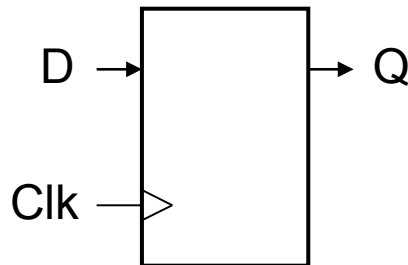
- Arithmetic/Logic Unit

- $Y = F(A, B)$



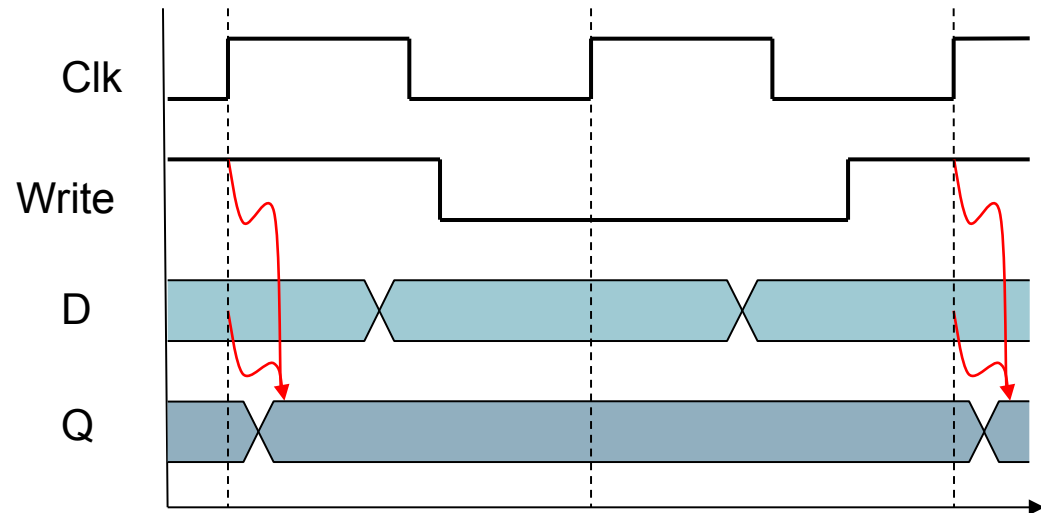
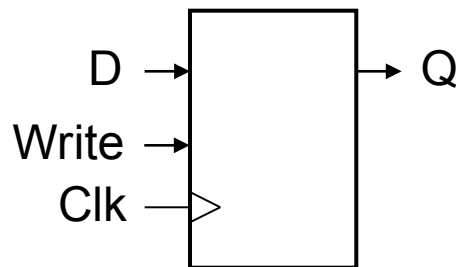
Sequential Elements

- Register: stores data in a circuit
 - Uses a clock signal to determine when to update the stored value
 - Edge-triggered: update when Clk changes from 0 to 1



Sequential Elements

- Register with write control
 - Only updates on clock edge when write control input is 1
 - Used when stored value is required later



Clocking Methodology

- Combinational logic transforms data during clock cycles
 - Between clock edges
 - Input from state elements, output to state element
 - Longest delay determines clock period

